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Accomplished

1. Detect Black Keys
2. Hand detection
 - a. Media pipe
3. Hovered Black Keys
 - a. Using values from the two previous parts

To be fixed

1. Optimize the algo (slow on detecting hovered black keys) where issues may possibly come from three places:
 - a. The way we loop through each piano coordinate and then loop through each fingertip coordinate
 - b. The constant instantiation of the set holding hovered notes
 - c. Updating the image of the piano
2. Centering and straightening the piano only works with static images, so during the demo, we used a version of the detect piano function without re-orientation and cropping. This means that our demo code assumes a piano is in the frame and is correctly orientated with ideal lighting. If this requirement is not met or there is no piano in the frame, the program may crash.
3. Black key detection is also currently hard-coded, starting with A# being the first black note to the left
4. When making a dictionary that will look like this
 Piano key (key): All possible coordinates (value)
 Since a possible test case for our piano is when the user takes a picture from the side of the piano, we will also see the sides of the key. We initially thought of using all possible coordinates and checking if the hand is in it to detect hovering. To optimize it, we can check for the 4 bound coordinates (top left, top right, bottom right, bottom left) and the slope from each point to the neighboring points. By doing this, we can calculate mathematically if our fingertip is hovering over a key instead of adding more runtime by making a set to hold all possible coordinates

Future Goals

1. Detect & Check which White keys are being hovered
2. Detect different types of pianos (not just rectangular ones)
3. Using 2 cameras and accurately detect when any key is being pressed

Explaining some files in the GitHub

(Test files are our main.py iterations, testing each part on static images in the images folder)

1. Test.py
 - a. Edge detection over the whole image
2. Test2.py
 - a. (tried) Detect the location of the piano and rotate the piano if it is not straightened
 - b. Edge detection
3. Test3.py
 - a. Draw and color the black keys on the piano
4. Main.py
 - a. Calls hand detection function
 - b. Crop the piano to avoid detecting other objects
 - c. Able to check if the hand is hovering over black keys
 - d. Take a picture immediately when the program is started and if no hand is detected in 10 seconds, retake the picture
5. PianoFuncions.py
 - a. File to hold helper functions for Main.py
 - b. Find and color the black notes from a greyscale image for labeling
 - c. Find all coordinates that relate to a certain note and put them in a dictionary
6. Utilities.py
 - a. Holds the function of drawing the hand-tracking annotations on the live feed

https://github.com/KymaiselHunter/CSCI_49369_Final_Project/tree/in_class_demo